Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 1 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

Number			
4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 1 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 1 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 1 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 1 of 7 · Question 4 part c · 2 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 2 of 7 · Question 2 part b · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

2(a)	Award **one** from each: Zafer: 1. Choose strong (1) authentication credentials (username/password) 2. Not sharing his credentials (1) 3. Not having automatic login to cloud / leaving his machine unattended (1) 4. Changing password regularly (1) 5. Password protect documents (1) Provider: 1. Infrastructure (e.g. firewall/servers) must be secure from unauthorised access (1) 2. Keeping their security software up to date (1) 3. Policy and procedures effective in preventing an insider attack / a data breach (1) 4. Data protection laws in the resident country must be obeyed (1) 5. Backup and restore procedures (1) 6. Encryption (1)		2

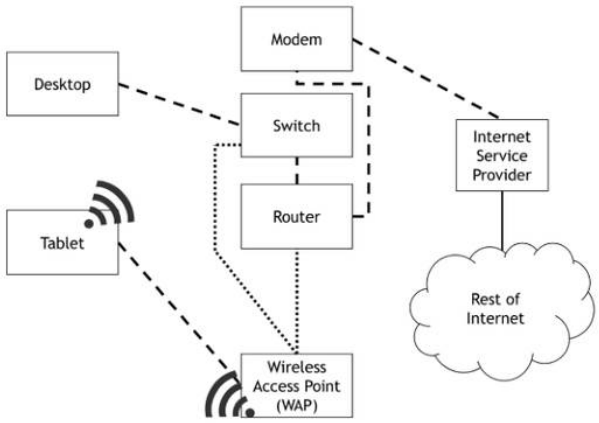
Question Number	Answer	Additional Guidance	Mark
2(b)	**Two** marks if completely correct. **One** mark for any two correct cells. **Robert** / **Zafer** Washing machine design drawings: Read / Read A folder of manuals for new machines: None / Read, Write Or Write, Read	Ignore spelling mistakes	2

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 2 of 7 · Question 2 part b · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)	Award **one** mark from each category (maximum of 2) Input: • www.pearson.com / website domain (1) • Uniform Resource Locator / URL (of Pearson website) (1) Output: • 2.20.38.113 (1) • Pearson's machine IP(v4) address (1)	Don't penalise spelling or typos, especially in addresses	2

Question Number	Answer	Additional Guidance	Mark
2(d)	One mark for each dotted connection, correctly located. Maximum of 6 marks. Note: WAP can be connected to either the Switch or the Router, but not both  		6

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 2 of 7 · Question 2 part b · 2 marks

2019 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 3 of 7 · Question 2 part a · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

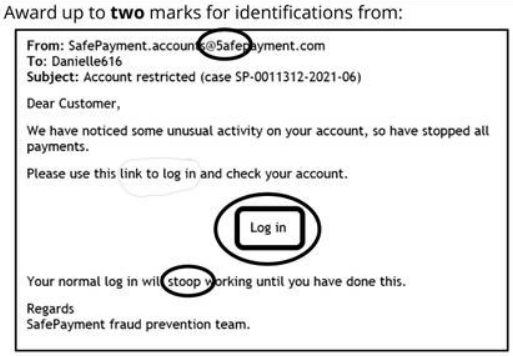
2(a)	**The only correct answer is B** *A is not correct because a password is needed to prevent unauthorised access to the network* *C is not correct because it only provides security after access to the network* *D is not correct because it only provides security after access to the network*		1

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 3 of 7 · Question 2 part a · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

2(b)(ii)	Award up to **two** marks for identifications from:  	Allow link to log in OR circle around button	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 3 of 7 · Question 2 part a · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 3 of 7 · Question 2 part a · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 4 of 7 · Question 1 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 5 of 7 · Question 4 part e · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

4(a)	**The only correct answer is C** ***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity* ***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity* ***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second***		1

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **one** mark for each correct cell: **Type** / **Example** / **Number of bits** IPv4 / 192.169.0.3 / 32 (1) IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1)		2

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 5 of 7 · Question 4 part e · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

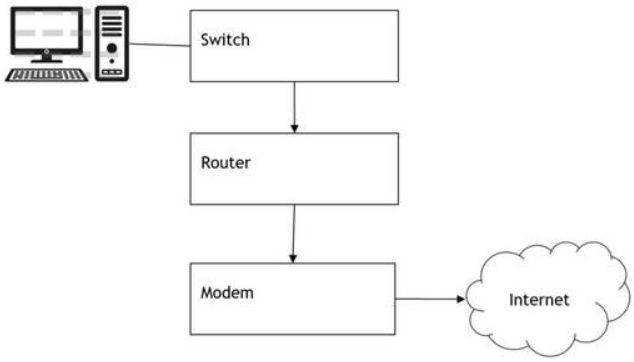
Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **two** marks for a linked description, such as: • In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1)	Allow radio signals, infra-red, microwave for air (must be more than just waves) Allow twisted pair, CAT for copper. Allow electrical/light signals (not just electricity)	2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. A protocol (suite/group/family) (1) 2. Used on a wired network / wired connection (1) 3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1) 4. Defines how packets are checked for errors (1) 5. Defines the speed of transmission (1) 6. Operates at the link layer of the TCP/IP stack (1)		2

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 5 of 7 · Question 4 part e · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award **three** marks for all of switch, router and modem in the correct order as in diagram Award **two** marks for all of switch, router and modem in any order Award **one** mark for any of switch or router or modem in any box 		3

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 5 of 7 · Question 4 part e · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(e)	Award up to **two** marks for a linked description, such as: Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1)	For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 6 of 7 · Question 1 part b · 1 mark

2024 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Number			
1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1) • WANs have lower bandwidth / LANs have higher bandwidth (1) • WANs connect multiple LANs / LANs are single networks (1) • WANs have higher latency / LANs have lower latency (1) • WANs use shared infrastructure / LANs are usually privately owned (1) • WANs use third party connections / LANs are owned by the organisation (1) • WANs have more users/devices / LANs have fewer users (1)	Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance. Two different differences must be identified. Do not accept general points such as WANs are slower - too vague. Do not accept LANs can be wireless.	2
1(b)	Award **one** mark for any of the following: • (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1) • (Increased risk of) hacking/unauthorised access/interception (1)		1
1(c)(i)	Award **one** mark for: • (Partially connected) mesh (1)		1

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 6 of 7 · Question 1 part b · 1 mark

2024 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

1c(ii)	Award **one** mark for any of the following: • More expensive to install (1) • Needs more cabling/wiring (1) • Requires more maintenance (1) • Harder to maintain / manage / install (1) • Higher power consumption (1)		1
1(c)(iii)	Award up to **two** marks for a linked explanation such as: • In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1)		2
1(d)	**The only correct answer is D** *A is not correct because the application layer deals with applications* *B is not correct because the transport layer deals with packet structure* *C is not correct because the network layer deals with routing end to end*		1
1(e)(i)	Award **one** mark for: • https/hypertext transfer protocol secure (1)		1
1(e)(ii)	Award **one** mark for: • specification(.html) (1)		1
		Total for question 1	**10**

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 6 of 7 · Question 1 part b · 1 mark

2024 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 7 of 7 · Question 4 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 7 of 7 · Question 4 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
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</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

Type MS · Keeping a network secure: access rights, physical security and authentication

Answer 7 of 7 · Question 4 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark